Research Statement

La Rosa Biagio

Deep learning has been the driving force behind recent advances in artificial intelligence (AI), resulting in widespread applications. Over the last decade, deep neural networks (DNNs) have grown in size and complexity, tackling tasks such as image recognition, game playing, and dialogue systems. Despite the mathematical transparency of the training process, the acquired knowledge and the rationale behind DNN predictions remain elusive. The lack of semantics encoded by DNNs and their opaque decision processes pose challenges in the development, deployment, and regulation of these systems [7]. My research aims to enhance our scientific understanding of deep learning, making its development and testing more controllable and verifiable. Specifically, I seek to uncover the **knowledge learned by DNNs** and explore designs that **improve their explainability**.

Probing Neurons

Probing neurons aims to extract knowledge acquired by neurons during the learning process within DNNs. This research area provides researchers with insights into the underlying mechanisms governing DNN design. My research delves into **metrics** designed to compare various approaches and investigates methods that contribute to **a more comprehensive understanding of neuronal behavior**. I argue that looking only at the alignment between activations and concepts is limiting. To address this limitation, my work suggests considering additional factors such as the frequency of detection, coverage of activations, and the correlation between activation values and the presence of a concept [8]. To gain deeper insights into neuronal behavior, my research demonstrates that **neurons can recognize different concepts at various activation values** (fig. 2). Consequently, approaches concentrating solely on a subset of activations, such as the average or the highest values, fail to capture the full breadth of information encoded in neural activations [8]. Therefore, I propose a novel methodology leveraging clustering algorithms and heuristic search to

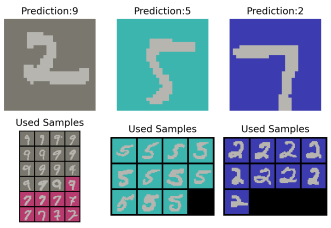

Figure 1: Examples of prediction and explanations by examples returned by memory augmented neural networks. By inspecting the memory, it is evident the bias learned by the network.

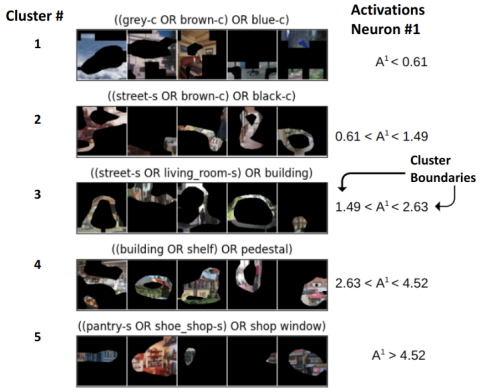

Figure 2: Concepts composition captured at different activation ranges (right side) by neuron #1.

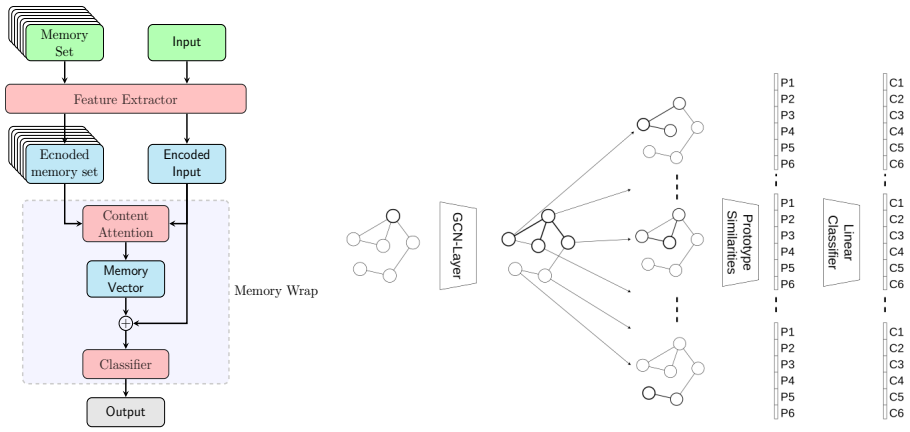

Figure 3: Examples of self-explainable neural networks. (Left) Memory Wrap combines the input and the memory into a memory vector based on their attention and computes the output. (Right) A prototype layer takes as input the graph's latent representation and returns the similarity with a set of learned prototypes. The classifier then computes the prediction based on the similarities.

detect the composition of concepts recognized at different activation ranges, thereby shedding light on neural phenomena like **progressive specialization** and **activation polisemy**.

Self-Explainable Deep Neural Networks

Self-explainable DNNs are a class of architecture designed to be partially interpreted via inspection [2, 1, 3]. My research endeavors focus on approaches of this class that modify or **incorporate elements into current state-of-the-art DNNs**, specifically targeting the last part of the network. Within this paradigm, I explore the integration of memory, prototypes, and constraints as elements aimed at enhancing interpretability, affording explanations linked to the decision-making process and computationally efficient. The underlying idea is to treat the layers preceding the classifier as a black-box feature extractor, transforming input into a representation capturing feature relationships. Then, the aforementioned elements are placed between the feature extractor and the classifier. My research targets designs where the classifier and the added elements are **independent of the feature extractor**, allowing for versatility across DNN architectures. This approach addresses a common limitation of self-explainable DNNs: constrained generalizability across different architectures and lower performance compared to black-box models.

Memory One of the elements explored in my research is the incorporation of a *memory* — an external module where the feature extractor can access and store information, mirroring the CPU-RAM communication paradigm in modern computers. The memory module facilitates the storage and utilization of crucial information during the decision process. My research proposes **tracking mechanisms of the writing and reading processes to provide interpretability**. The contribution lies in diverse methodologies [5, 6] to embed a memory module within a network (e.g., fig. 3) and to generate various explanations like feature attribution, explanations by examples (fig. fig. 1), and counterfactuals. The efficacy of memory tracking is validated across different architectures (i.e., recurrent, convolutional, and attention-based ones) and tasks. Remarkably, DNNs augmented with a memory, and optimized for interpretability, do not lower their accuracy; instead, they can exceed the performance of their DNNs counterpart when trained on a small set of data [6].

Prototypes Prototypes are vectors learned and adjusted during the training process to represent sets of data or concepts. When an input is fed to the network, it activates the prototypes at different strengths. My research [11] showcases the application of **prototypes-based layers on graph data** (Figure 3 (right)). In this case, a prototype represents a subgraph generated by a node and its neighbors. We found that by attaching prototypes-based layers to several graph neural networks (i.e., GCN, GAT, GIN), we can improve their performance and reach, at the same time, fidelity in explanations comparable to state-of-the-art XAI approaches.

Contraints *Contraints* are biases introduced in the training process to maximize interpretability. For example, in prototypes-based layers, we use constraints to ensure diversity among prototypes [11]. My research proposes constraints to **align the latent space axis with the structural components of molecules**. We show that the inspection of the axes explains how properties in molecular regions contribute to modulating biological behavior. Moreover, networks trained with these constraints reach better (or equal) performance with respect to their black-box versions [10].

Future directions

Aligned with my long-term objective of elucidating the inner workings of DNNs through an exploration of their elements, I propose to delve deeper into probing neurons and subsequently integrate the acquired insights into self-explanatory DNNs.

In the immediate future, I will scrutinize the **optimality of existing neural explanations**. This involves an examination of the effectiveness of current approaches and the identification of the challenges connected to the optimality. Simultaneously, I plan to explore **designs capable of leveraging the recognition of concepts and prototypes at different activation levels**. This exploration seeks to identify novel design paradigms that may offer enhanced interpretability without compromising performance.

Moving towards the long term, a critical objective is to render the probing process independent of concept datasets. While some endeavors propose using families of multi-modal architectures to replace datasets [4, 9], my goal is to achieve complete **independence from external factors**. This aims to mitigate biases and ensure the reliability and generalizability of the probing process. In pursuit of this independence, I will investigate the application of methods rooted in pattern recognition. This includes the exploration of techniques to detect similarities in both activations and inputs. The employment of novel clustering algorithms will complement this effort, providing a more robust foundation for the subsequent phases of the research. Based on the outcomes of the above investigations, the ultimate objective is to identify differentiable mechanisms capable of direct **integration into self-explanatory DNNs**. This step is crucial for incorporating the acquired knowledge into practical and interpretable model architectures.

Another path to explore to understand knowledge learned by the network is the **relations encoded in the weights** connecting neurons. The intuition is that, given neurons' role as detectors of concepts and the weights representing the learned parameters, the optimization process during training could encode relationships directly within the weights. Shedding light on the relationships embedded in weights contributes to a deeper understanding of the mechanisms governing the training process itself and provides valuable insights into the optimization strategies employed.

In summary, this research agenda aims to discover the knowledge learned by DNNs and explore designs that improve their explainability, contributing to the ongoing discourse on deep neural network interpretability.

References

[1] D. Alvarez Melis and T. Jaakkola. Towards robust interpretability with self-explaining neural networks. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018. URL `https://proceedings.neurips.cc/paper_files/paper/2018/file/3e9f0fc9b2f89e043bc6233994dfcf76-Paper.pdf`.

[2] C.-H. Chang, R. Caruana, and A. Goldenberg. NODE-GAM: Neural generalized additive model for interpretable deep learning. In *International Conference on Learning Representations*, 2022. URL `https://openreview.net/forum?id=g8NJR6fCCl8`.

[3] C. Chen, O. Li, D. Tao, A. Barnett, C. Rudin, and J. K. Su. This looks like that: deep learning for interpretable image recognition. *Advances in neural information processing systems*, 32, 2019.

[4] E. Hernandez, S. Schwettmann, D. Bau, T. Bagashvili, A. Torralba, and J. Andreas. Natural language descriptions of deep visual features. In *International Conference on Learning Representations*, 2022. URL `https://arxiv.org/abs/2201.11114`.

[5] B. La Rosa, R. Capobianco, and D. Nardi. Explainable inference on sequential data via memory-tracking. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI-20*, pages 2006–2013. International Joint Conferences on Artificial Intelligence Organization, 7 2020. doi: 10.24963/ijcai.2020/278. URL `https://doi.org/10.24963/ijcai.2020/278`.

[6] B. La Rosa, R. Capobianco, and D. Nardi. A self-interpretable module for deep image classification on small data. *Applied Intelligence*, aug 2022. doi: 10.1007/s10489-022-03886-6.

[7] B. La Rosa, G. Blasilli, R. Bourqui, D. Auber, G. Santucci, R. Capobianco, E. Bertini, R. Giot, and M. Angelini. State of the art of visual analytics for explainable deep learning. *Computer Graphics Forum*, 42(1):319–355, Feb. 2023. ISSN 1467-8659. doi: 10.1111/cgf.14733.

[8] B. La Rosa, L. H. Gilpin, and R. Capobianco. Towards a fuller understanding of neurons with clustered compositional explanations. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. URL `https://openreview.net/forum?id=51PLYhMFWz`.

[9] T. Oikarinen and T.-W. Weng. CLIP-dissect: Automatic description of neuron representations in deep vision networks. In *The Eleventh International Conference on Learning Representations*, 2023. URL `https://openreview.net/forum?id=iPWiwWHc1V`.

[10] M. Proietti, A. Ragno, B. La Rosa, R. Ragno, and R. Capobianco. Explainable AI in drug discovery: self-interpretable graph neural network for molecular property prediction using concept whitening. *Machine Learning*, oct 2023. doi: 10.1007/s10994-023-06369-y.

[11] A. Ragno, B. La Rosa, and R. Capobianco. Prototype-based interpretable graph neural networks. *IEEE Transactions on Artificial Intelligence*, pages 1–11, 2022. doi: 10.1109/TAI.2022.3222618.